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DNA barcoding of more than one million insect specimens indicates that at least one third of California's insect biodiversity remains undiscovered

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Insects are the most diverse terrestrial organisms, yet they are underrepresented in large-scale conservation assessments. To address this gap, the California Insect Barcoding Initiative is developing a publicly accessible, statewide DNA-barcode reference library for California insects to support scalable surveys, establish baseline biodiversity measurements, and generate potential distribution maps that inform conservation planning. Specimens are collected with a hybrid strategy that combines standardized Malaise trapping, to enable replicable sampling, and opportunistic collecting, to maximize taxonomic coverage. To date, we have barcoded over one million specimens; preliminary completeness analyses suggest that current sampling captures roughly 65–69% of the fauna, implying a conservative minimum of circa 61,000 insect species in California. Using all sequenced specimen records, we generated rule-based spatial range interpolations constrained by ecoregion and vegetation type, and used these to infer spatial patterns in insect species richness across the state. We identify areas of both high and low potential species richness, with current peaks in the Southern California Mountains ecoregion and the Mojave Basin and Range ecoregion. Our species richness estimates and spatial patterns are explicitly provisional and are expected to evolve as sampling gaps are addressed. Finally, we make all sequence data, specimen images, and occurrence records publicly available via the Barcode of Life Datasystem. This ongoing effort constitutes the first large-scale DNA barcode-based survey for California insects, providing an expandable foundation for tracking temporal change, testing drivers of insect diversity, and prioritizing regions for conservation and targeted inventory.

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Introduction

Our planet is losing species at an unprecedented rate, which many scientists suggest may be an anthropogenic mass extinction event (Cowie *et al*., 2022). This includes losses in one of the most diverse and abundant groups of organisms on the planet – the insects (Wagner *et al*., 2021). In most terrestrial ecosystems insects represent the largest proportion of animal diversity. Early estimates of insect biodiversity suggested that there are approximately 30 million insect species worldwide (Erwin, 1983), while more recent estimates converge around 6 million species (Basset *et al*., 2012; Novotny *et al*., 2002; Stork *et al*., 2015). New cryptic species revealed by DNA sequencing further increase these estimates to circa 20–40 million insect species worldwide (Larsen *et al*., 2017; Li & Wiens, 2023). Insects' importance for maintaining healthy food webs and ecosystem services (e.g., pollination, pest control, dung burial) was valued at $57 billion annually in the USA in 2006 (Losey and Vaughan, 2006). When adjusted for inflation, this figure is equivalent to $109.1 billion in 2025 (Bureau of Labor Statistics, n.d.). Insects' tremendous diversity and provisioning of critical ecosystem services make their ongoing, severe population declines extremely troubling (Dalton *et al*., 2023; Hallman *et al*., 2017; Janzen & Hallwachs, 2021; Shortall *et al*., 2009).

Despite the ecological importance of insects and evidence of their global decline, critical data for insect distributions and population trends are lacking for most areas. This is problematic for conservation efforts, as distribution maps are essential for spatial conservation planning (Porfirio *et al*., 2014). For instance, conservation practitioners in California use the ACE 3.0 Terrestrial Biodiversity Summary of California maps to guide decision making (California Department of Fish and Wildlife, 2018); however, these maps only incorporate data from birds, mammals, amphibians, reptiles, and plants, which may serve as poor proxies for insects and other invertebrates (Oberprieler *et al*., 2019). Moreover, to date most research on insect distributions, diversity, and sampling completeness has focused on a limited subset of lineages, primarily charismatic groups like bees (Chesshire *et al*., 2023; Orr *et al*., 2021) and butterflies

(Sánchez-Fernández *et al*., 2021; Shirey *et al*., 2021), leaving many major taxonomic groups a mystery. Addressing these data gaps requires large-scale insect surveys, development of species checklists, and methods for documenting and delimiting unknown species.

Understanding insect diversity patterns is valuable not only for their conservation, but also because several aspects of their biology make insects an excellent model system for monitoring broader, long-term changes in community biodiversity. For instance, many insects are highly endemic and sensitive to changing environments (Parmesan *et al*., 1999), and many species can serve as indicators of shifts in local seasonality, given their dependence on temperature for development (Forrest, 2016; Roy & Sparks, 2020). Moreover, insects exhibit higher total species richness and abundance than vertebrate groups (Mora *et al*., 2011; Rosenberg *et al*., 2023), facilitating more precise comparisons of diversity across sites and over time. Insects are also relatively easy to passively sample compared to vertebrates, which often require capture-and-release or field observations at the right time and place (Sutherland, 2006). Consequently, insect biodiversity is already used as a staple of assessing the health of freshwater ecosystems (e.g., the Ephemeroptera, Plecoptera And Trichoptera (EPT) Index; Blachuta *et al*., 2014).

Despite these advantageous characteristics, insects are rarely, if ever, used in terrestrial biodiversity estimates for conservation planning (Nania *et al*., 2023). Moreover, significant monitoring gaps exist for insects, largely because identifying large numbers of specimens, collected or observed, to the species level is difficult, time-intensive, and can require high levels of taxonomic expertise. Only a few efforts are being made in the United States to perform long-term insect population monitoring at scale, including the National Ecological Observatory Network (NEON; Thorpe *et al*., 2016) and the Long-Term Ecological Research (LTER; Crossley *et al*., 2020) sites. While these programs represent unique opportunities to monitor insect biodiversity across the country, they have focused on specific groups of insects, largely determined by collection methods used (e.g., Coleoptera using pitfall traps or mosquitoes using carbon dioxide traps) (Crossley *et al*., 2020; Thibault *et al*., 2023). More systematic approaches for monitoring insects are needed to improve the efficiency (in terms of cost and quantity) of both collecting and identifying vast amounts of insect specimens.

DNA barcoding is a rapid and affordable approach that has become widely used in the past several decades for identifying and even delimiting species based on the sequences of short segments of DNA (e.g. the *cytochrome c oxidase subunit I* (*COI*) locus for animals; Hebert *et al*., 2003; Wilson, 2012). DNA barcoding approaches have enabled insect biodiversity studies to extend beyond groups that are readily identifiable from morphology alone (e.g., butterflies, bumblebees) to include groups that are highly diverse but largely undescribed (often referred to as "dark taxa", e.g., many small flies and wasps) (Meier *et al*., 2025). DNA-based observatories are increasingly recognized as essential infrastructure for scalable biodiversity monitoring and forecasting, integrating barcode data with ecological models and global survey networks (Ovaskainen et al., 2025).

Several major regional efforts have developed large-scale insect inventories using DNA barcoding. In the Área de Conservación Guanacaste (ACG) in northwestern Costa Rica, taxonomists have undertaken insect sampling since 1978 and implemented voucher-based DNA barcoding starting in 2003 (Janzen & Hallwachs, 2011). This effort formed the basis for the All Taxa Biodiversity Inventory (ATBI) concept (Janzen & Hallwachs, 1994). The ACG dataset primarily consists of reared caterpillars and their parasitoids, providing valuable data on multi-trophic interactions and on a highly species-rich dark taxon, parasitoid insects. The combination of these reared samples with an extensive series of Malaise traps allows the two data sources to inform each other, especially for understanding completeness of the ATBI efforts (Quicke *et al*., 2024; Colwell *et al*., in review). A survey in the Southern Appalachian Sky Islands leveraged DNA barcoding of specimens from leaf litter samples to investigate community composition and local endemicity (Caterino & Recuero, 2023). In Canada, a multi-method approach has been used to DNA barcode one million insect specimens from across the country, with the ultimate goal of developing more accurate planetary species richness estimations (Hebert *et al*., 2016). Many areas around the world would benefit from similar sampling efforts, and a key priority for the research and conservation communities should be to extend surveying of insects in areas identified as biodiversity hotspots (Myers *et al*., 2000).

The California Floristic Province is one such hotspot. It is the most biodiverse region in North America (Calsbeek *et al*., 2003), and is considered to be in the world's top 25 most biologically rich and threatened terrestrial ecosystems (Myers *et al*., 2000). Roughly 87% of this biodiversity hotspot occurs within the state of California, and it covers approximately 70% of the state's land area.

The California Floristic Province contains ~20% of all vascular plant species known from the United States and more than 30% of known insect species north of Mexico (Calsbeek *et al*., 2003; Franklin *et al*., 2021). Though the state of California harbors substantial insect biodiversity, most insect taxa lack a comprehensive checklist of species present in the state, a data gap that is particularly acute for the most diverse groups with high numbers of dark taxa (e.g., Diptera, parasitic Hymenoptera). As a result, the true proportion of North American insect species found in California remains unknown, hampering conservation efforts.

Efforts to understand and conserve California's biodiversity require more complete knowledge of the state's insect fauna. In alignment with the 2030 targets of the Kunming-Montreal Global Biodiversity Framework (Convention on Biological Diversity, 2022), California has committed to preserving 30% of land within the state by 2030 (California Natural Resource Agency, 2021). Achieving these targets requires identifying the distribution of available conservable land in California, coupled with clarifying the spatial distribution of biodiversity in the state. Without insect distribution maps, we lack critical information needed to incorporate these important species within this framework. Moreover, detecting the effects of climate change and biodiversity loss requires monitoring to begin immediately to establish a baseline and continue into the future.

To facilitate these goals, natural history museums, research institutions, and other partners across California have come together to collect and sequence DNA barcodes for every insect species present in the state. This state-funded collaborative project, called the California Insect Barcoding Initiative, is part of a larger all-taxon biodiversity inventory effort in California, CalATBI. The project's primary objectives are to establish a voucher-based DNA barcode reference library for California insects to improve the efficiency of future monitoring, provide a biodiversity baseline for future sampling comparisons, generate potential range maps for insect biodiversity that can inform conservation metrics, and preserve newly-collected specimens for future study, while rescuing at-risk specimen collections from degradation or disposal.

This project's methodology was designed to maximize the biodiversity sampled across space and time, generating a dataset that can be used to catalog species, discover new taxa, track phenology, strategically target undersampled areas, and explore patterns of insect biodiversity. Development of a DNA barcode reference library requires sampling authoritatively identified specimens from legacy collections, coupled with making new collections in the field to capture species that are unknown or unrepresented in collections. The present study focuses solely on the new field collection data, prior to the integration of identified specimens from legacy collections; therefore, we rely on molecular operational taxonomic units (MOTUs) as a proxy for species. The goals of the present study are to:

i) assess statewide species richness and sampling completeness across the project using Chao species richness estimators;

ii) infer statewide patterns of insect community dissimilarly;

iii) use rule-based spatial range interpolations to generate potential range maps for all insect species collected in California, and

iv) use these predictions to infer spatial patterns in potential insect biodiversity across the state.

This dataset represents one of the largest and most diverse insect barcoding datasets to date, with 1,030,728 sequenced specimens belonging to 42,305 unique MOTUs. This dataset facilitates the assessment of current sampling completeness and preliminary assessments of broadscale insect biodiversity patterns across California. The California Insect Barcoding Initiative is laying the groundwork for future insect surveys and monitoring programs by developing a DNA barcode reference library to enable rapid identifications, developing a methodology that can be replicated to track phenology or serve as a model for other insect biodiversity surveys, and developing distribution and biodiversity mapping protocols that support targeted resampling and conservation planning.

Methods

Data collection overview

One of the primary goals of the California Insect Barcoding Initiative was to create a comprehensive reference library of DNA barcodes for all hexapod species (insects, collembolans, proturans, and diplurans; henceforth referred to as insects) of California, including both known species and those new to science. To facilitate this goal, we maximized both the spatial and temporal coverage of our field collecting by sampling across as many ecoregions as possible and by revisiting locations throughout the year. We used a two-pronged approach based on i) **Observatory Sites**, locations where standardized collecting was conducted by running Malaise traps continuously, and ii) **Snapshot Sites**, locations sampled at single time points using a wide variety of collecting methods. Snapshot Sites were selected to cover as many different ecoregions as possible, maximizing coverage of distinct plant communities and abiotic factors (Environmental Protection Agency Level IV Ecoregions; Griffith *et al.*, 2016). Most samples were collected fresh for this project starting in 2022, aside from a small number of samples that were retrieved from cold storage dating back to 2016. Achieving this large-scale sampling effort required a multi-institutional collaborative approach, with participation from several natural history museums, universities, government agencies, and other groups (Supplementary Table 1).

Observatory Sites

Observatory Site collections (Supplementary Table 2) used Townes-style Malaise traps (Townes, 1972) purchased from Sante Traps (Lexington, Kentucky). Traps typically operated for one-week intervals followed by sample pickup, unless circumstances like collector unavailability rendered weekly sampling infeasible. Insects were collected into 95% ethanol, and samples were stored in a cool dry place (freezers or refrigerators when available) until batches of samples were retrieved by California Insect Barcoding Initiative field crews for storage in freezers.

The standardized approach used at Observatory Sites generated abundance data in addition to presence data, producing baseline estimates of both species diversity and abundance to facilitate future efforts to monitor insect communities. Many Malaise traps were deployed near weather stations, allowing factors like precipitation, temperature, and humidity to be correlated with insect phenology. In total, 74 Observatory Site locations are included in the dataset, representing 13 of 14 Level III EPA ecoregions in California (Fig. 2A).

Snapshot Sites

Snapshot Site collecting approaches were not standardized, as collectors needed the flexibility to collect for variable durations as deemed necessary, with methods optimized for different environments and for particular target insect taxa. Methods comprised sweeping and beating vegetation, hand collecting, short-term Malaise trapping, pan trapping, blue vane trapping, pitfall trapping, blacklight trapping, mercury vapor light trapping, sifting leaf litter, dip netting, Lindgren funnel trapping, and Berlese and Winkler extractions of soil and leaf litter.

Insects were generally collected directly into 95% ethanol (or transferred into 95% ethanol from soapy water in the case of pan traps) and were then stored in a freezer prior to sequencing. In total, more than 500 Snapshot Site locations are included in this dataset, representing collections from all Level III EPA ecoregions in California (Fig. 2A).

Sample processing

For Observatory Site samples, we recorded samples' wet weight (Supplementary Table 1) following the protocols used by the Centre for Biodiversity Genomics (University of Guelph, Guelph, Ontario, Canada). After weighing, some individuals were removed prior to plating and sequencing (duplicate specimens of eusocial insects like honeybees and ants, and excess

Collembola). Groups that we expect to represent dark taxa with high potential for species discovery were included in full for sequencing. For every Observatory Site location, one sample per month was submitted for sequencing at the Centre for Biodiversity Genomics; this was typically the sample corresponding to the first full week of collecting in that month. Additional Observatory Site samples which were not immediately sent for sequencing are being stored in freezers at the collection institutions.

Snapshot Site samples were not weighed, and were pre-sorted prior to sequencing by removing duplicate specimens within morphospecies, so as to reduce sequencing redundancy and to voucher additional specimens in institutional collections. All sorted samples were then sent to the Centre for Biodiversity Genomics for sequencing following the same protocols for Observatory Site samples.

Use of Barcode Index Numbers for species delimitation and identification

We chose to use the Centre for Biodiversity Genomics' Barcode Index Numbers (BINs) to identify the specimens collected through the California Insect Barcoding Initiative. Pairing BINs with a well-developed reference library can result in rapid species-level identifications from a single inexpensive and easily sequenced gene region, and even in the absence of a reference library, BINs can themselves be used as functional molecular operational taxonomic units (MOTUs). Although not a perfect proxy for species, the number of BINs estimated from large biodiversity datasets is strongly correlated with the number of morphologically identified species in those datasets (Salis *et al*., 2024). In the absence of comprehensive reference libraries, which do not yet exist for insects, approximating species using BINs or other molecular delimitation methods is the only practical approach for assessing patterns of biodiversity in lineages like insects that contain many dark taxa.

The Centre for Biodiversity Genomics provides a framework for large-scale DNA barcoding projects, from multiplex sequencing, to data storage on Barcode of Life Datasystems

(BOLD; Ratnasingham & Hebert, 2007), to assigning BINs to specimen records (Ratnasingham & Hebert, 2013). The specimen processing, sequencing, and bioinformatic methods in the present study follow Hebert *et al*. (2018) and deWaard *et al*. (2019) and are summarized below.

Specimen preparation and imaging

At the Centre for Biodiversity Genomics, specimens were size-sorted. Individuals smaller than 5 millimeters were arrayed directly into 96-well microplates, while specimens larger than 5 millimeters were either pinned, placed in a matrix tube, placed in a scintillation vial, or placed in an envelope as appropriate for their taxonomy and then arrayed in a 96-grid Schmidt box. High resolution images were taken of each specimen, with specimens in microplates imaged using a Keyence VHX System (Steinke *et al*., 2024b), and dried, pinned specimens imaged using an Automatic Imaging Rig (Steinke *et al*., 2024a). Images were associated with specimen records and assigned an Order-level identification using a machine learning algorithm (Steinke *et al*., 2024b). For larger pinned specimens, a small tissue fragment (typically a leg) was then removed and placed into a microplate, and the remaining portion of the specimen was labelled and placed into the Centre for Biodiversity Genomics' Specimen Archive.

DNA extraction and sequencing

An automated, magnetic bead-based DNA extraction was performed on all of the microplates containing specimens or tissue. Tissue lysis was performed overnight at 56°C before DNA purification. Following purification, all plates containing whole voucher specimens <5mm were deposited in the Centre for Biodiversity Genomics' Specimen Archive. Aliquots of DNA extracts from the microplates were consolidated into 384-well PCR plates for amplification, and remaining extracts were stored in a -80°C freezer archive. Unique Molecular Identifiers (UMIs) were incorporated into the amplicons to map sequence reads back to specimens, and PCR amplification targeted the cytochrome c oxidase subunit I (COI) gene region. Amplicons from 24 PCR plates were pooled on a PacBio Single-Molecule Real Time (SMRT) cell for sequencing. SMRT cells were pooled prior to annealing the sequencing primer and polymerase to create the final sequencing library, which was loaded onto a SMRT cell chip and sequenced on a Sequel IIe Sequencing Platform.

Following sequencing, a high-quality consensus sequence was generated during Circular Consensus Sequencing (CCS) analysis. Reads were de-multiplexed and mapped to the SMRT cell and source specimen using the SMRT cell adapter and UMI sequences, respectively.

Taxon delimitation

Consensus COI sequences were assigned to specimens and validated internally at the order level before the BOLD Refined Single Linkage (RESL) algorithm assigned BINs to specimen records. Taxonomic resolution was further improved using matches to previously sequenced BINs, sequence proximity in neighbor-joining trees, digital morphology, or via further examination by expert taxonomists (Hebert *et al*. 2018; deWaard *et al*. 2019).

Data cleaning

Specimen data from all subprojects funded by the California Insect Barcoding Initiative were obtained from BOLD on June 26, 2025. Non-hexapod data was filtered out, resulting in 1,030,728 specimen records. Records without assigned BINs were then removed, resulting in 892,274 records for downstream analysis. These records comprise 42,305 unique BINs (Supplementary Table 3).

In some instances, a single BIN had multiple, conflicting taxonomic assignments. To address these cases, we cleaned and standardized specimen record data. If some records within a BIN had taxonomy only partially assigned (e.g., with family as the most-specific assignment) while other records of that same BIN had a more specific taxonomic identification (e.g., species as the most-specific assignment), all records for that BIN were assigned to the most specific rank possible. If records within a BIN had multiple, conflicting taxa assigned at a given rank (e.g., some records were identified as the family "Cicadellidae" while others were identified as "Cercopidae"), assignment was determined manually based on a combination of (1) whether either each identification had an expert identifier or had been auto-assigned, (2) examining the specimen images on BOLD, and (3) comparing the number of records corresponding to each taxon assignment; in most cases, there were few erroneous assignments mixed in with a large majority of accurate assignments. Hereafter, we refer to BINs as species, while recognizing that they most accurately reflect MOTUs. Data was also cleaned to ensure that all latitude and longitude values for a given collecting event were identical (removing rounding errors) by choosing the values that were assigned to the greatest number of records for a given event.

To better understand how the data generated through the California Insect Barcoding Initiative has contributed to our understanding of the state's insect fauna, insect records were downloaded from the Global Biodiversity Information Facility (GBIF) on August 21, 2025. These data comprised records derived from material samples and preserved specimen records within California for Insecta (GBIF.org, 2025a), Collembola (GBIF.org, 2025b), Diplura (GBIF.org, 2025c), and Protura (GBIF.org, 2025d).

Identifying taxonomic patterns

To assess specificity of specimen identifications, we calculated the proportion of specimens identified to the Class, Order, Family, Subfamily, Genus and species levels. We also calculated the number of species collected for each order in our dataset. For the GBIF dataset, we calculated the number of species per order after filtering the data for records where the taxonomic rank was species and the taxonomic status was accepted (Supplementary Table 4).

To understand patterns of species abundance, we generated rank-abundance curves for all species in our dataset.

Assessing the impact of collecting methods

Collecting methods substantially influence the taxonomic composition of insect samples; in particular, Malaise traps are biased towards capturing flying insects. To quantify these patterns, we calculated the fraction of species that were collected only via Malaise traps, only via non-Malaise trap methods, or via both sets of methods. We also calculated the Order-level taxonomic composition for Malaise and non-Malaise samples.

Assessing seasonal patterns in collecting

We assigned each collecting event and each individual specimen to a season. Seasons were defined as follows: spring (March–May), summer (June–August), fall (September–November), and winter (December–February). We then calculated the percent of collecting events and specimens for each season.

Estimating collecting bias across ecoregions

To assess the degree to which our sampling effort was even across ecoregions, we calculated a collecting bias metric as follows:

$$Collecting\ bias = \frac{Number\ of\ collecting\ events\ in\ the\ ecoregion}{Total\ number\ of\ collecting\ events} \Big/ \frac{Area\ of\ the\ ecoregion}{Total\ area\ of\ California}$$

For ecoregions extending outside of California, we only considered the area of the ecoregions within California. We then compared collecting bias for each ecoregion with the total number of unique BINs collected in each ecoregion.

Predicting insect biodiversity and estimating per-site sampling completeness

Despite significant sampling effort, it is highly unlikely that we sampled every insect species present at each collecting site. We used three methods to estimate a lower bound for the number of species present in California: using rarefaction and extrapolation with the Chao1 estimator, implemented in the R package `iNext` for all samples in the dataset; using the iChao1 estimator, implemented in the R package `SpadeR1, for all samples in the dataset; and using the iChao1 estimator for only Observatory Site samples in the dataset (Fig. 3A; Chao *et al*. 2009; Chao *et al*. 2016; Hsieh *et al*. 2016).

Each of these methods has different strengths and weaknesses. The Chao1 estimator underestimates species richness relative to the iChao1 estimator; however, using this estimator as implemented in `iNext` allowed us to estimate the total number of species present in California via extrapolation (Hsieh *et al*. 2016). Because the iChao1 estimator is sensitive to the number of uncommon species (Chiu *et al*. 2014), we expect that implementing this estimator for the Observatory and Snapshot Sites in combination will produce an overestimate of species richness, as targeted collecting and sorting Snapshot Site samples results in an increased number of rare species. However, implementing it for Observatory Sites alone will produce an underestimate, as the total number of specimens to use in the inference is necessarily lower.

For comparison, we additionally generated rarefaction curves for (1) all samples statewide and (2) only Observatory Site samples.

To understand how patterns of sampling completeness varied across the landscape, and because our long-term Observatory Site Malaise traps do not have collecting biases related to species rarity, we also predicted the total number of species present at Observatory Sites using the iChao estimator (Fig. 3B; Chiu *et al*. 2014). We obtained the iChao1 estimator for the subset of Observatory Sites with (1) ten or more sequenced samples, and (2) greater than 80% of taxa receiving a BIN assignment. We then calculated per-site sampling completeness estimates using the ratio of observed species to predicted species at a given site (Fig. 3C).

Measuring site and ecoregion dissimilarity

To understand the extent to which ecoregions are useful proxies for distinct insect assemblages, we first calculated the mean dissimilarity for each pair of EPA Level III ecoregions. To do so, we calculated the pairwise Bray-Curtis dissimilarity index for all pairs of Observatory Sites in our dataset, using the function `vegdist` from the R package `vegan` (Oksanen *et al.* 2013). We then calculated mean dissimilarity for sites from each pair of ecoregions. Second, we tested whether Observatory Sites located in (1) the same EPA Level III ecoregion have more similar insect communities than Observatory Sites located in either (2) different and adjacent, or (3) different and nonadjacent ecoregions. We classified pairs of ecoregions as adjacent or nonadjacent using the function `st_intersects` from the R package `sf` (Pebesma & Bivand 3). We then utilized Bayesian multilevel zero/one inflated beta regression to assess differences in mean pairwise Bray-Curtis dissimilarity across the three categories of site pairs (same ecoregion, adjacent ecoregions, nonadjacent ecoregions) while accounting for month of sampling and site identity. For these analyses, Observatory Site Malaise samples were used as replicates, and sampling month and site were included as random factors. Models were run with specifying a zero/one inflated Beta regression, with 4 chains, 7 cores, 20,000 iterations, and a maximum tree depth of 20 in the `brms` package using RStan (Bürkner, 2017).

Inferring the potential distribution of insect biodiversity across California

We inferred potential distributions by projecting observed occurrences onto combinations of ecoregions and vegetation types where taxa were sampled, generating habitat-constrained range estimates at 1-km resolution. For each species, rule-based spatial range maps were generated following a modified version of the approach used by Pinkert *et al*. (2023) and Chesshire *et al*. (2023). We chose this approach over the minimum convex hull approach, because the latter method can only produce contiguous range maps, thus inflating the predicted biodiversity of Central California where most species ranges would artificially appear to overlap. Additionally, our approach permits the inclusion of species represented by as few as a single specimen because it does not rely on drawing polygons around multiple specimen records.

To generate these range maps, we associated each species with (1) the Level IV ecoregions where it occurred in our dataset, using EPA ecoregion shapefiles (U.S. Environmental Protection Agency, 2024), and (2) the vegetation types where specimens were sampled, using Existing Vegetation Type (EVT) data accessed from the LANDFIRE database (LANDFIRE, 2022). Ecoregional maps were then filtered to only contain the LANDFIRE vegetation types where species were found. The EVT data are at a 30m resolution and our data had over 42,000 species; therefore, to improve computational speed, species range maps were inferred at a 1km resolution across California.

To understand aggregate patterns of sampled species richness across the state, we hierarchically combined the individual range maps into family-level, order-level, and all taxa maps (Fig. 4, Supplementary File 1). These maps display the potential estimated species richness across California at a one-kilometer resolution. Unlike traditional species distribution models, which statistically estimate environmental response functions, our approach uses observed occurrences combined with ecological stratification (ecoregion + vegetation type) to interpolate presence across unsampled locations. This method is scalable to tens of thousands of species without requiring environmental covariates or parametric inference.

Results

Collecting and sequencing results

We deployed Malaise traps at a total of 72 Observatory Sites. For clarification of terms, samples refer to bulk samples containing many specimens from a single collecting event, and specimens refer to the individual insects. Collecting at Observatory Sites resulted in between 2–102 bulk samples per site. For each Observatory Site, between 1–22 (mean = 7) bulk samples were fully sequenced at the Centre for Biodiversity Genomics, yielding a total of 840,260 sequenced specimens. Wet weights for samples are reported in Supplementary Table 2.

We collected at more than 500 Snapshot Sites using a variety of collecting methods, with many sites revisited for collecting multiple times. We sequenced 1,280 sorted samples derived from Snapshot Site collecting, yielding a total of 190,468 sequenced specimens.

Taxonomic patterns

The California Insect Barcoding Initiative project has sequenced 1,030,728 insect specimens as of June 25, 2025. Of these, 892,274 specimens have been assigned BINs, comprising 42,305 unique BINs or species. Most specimens have identifications at the family (38.4%), genus (24.2%) or species levels (26.5%), while fewer specimens have been identified only to order (1.7%) or class (<1.0%) (Fig. 1A). Additional specimens from this project continue to be submitted for sequencing and BIN assignment, and identifications will improve as additional reference specimens are sequenced and added to the BOLD barcode library.

In our sampling, the orders of insects with the highest levels of diversity are Diptera (16,181 species, comprising 38.2% of all sampled species), Hymenoptera (13,966 species, 33.0%), Coleoptera (4,234 species, 10.0%), Lepidoptera (3,126 species, 7.3%), and Hemiptera (2,934 species, 6.9%), with 1,864 species for all other orders combined (4.4%) (Fig. 1B). There are substantial differences between our dataset and the insect records for California present in the Global Biodiversity Information Facility (GBIF) database. The GBIF records comprise 17,854 species belonging to the following orders: Coleoptera (5,186 species; 29.1% of all species), Hymenoptera (3,611; 20.2%), Lepidoptera (3,520; 19.7%), Diptera (2,353; 13.2%), Hemiptera (1,568; 8.8%), and all others combined (1,616; 9.1%) (Fig. 1B).

The vast majority of species sequenced could be considered uncommon, given the low frequency with which they occur in our data, with 78.7% of species represented by fewer than 10 sequenced individuals (Fig. 1C). On the other end of the spectrum, 103 species were so abundant that 1,000 or more individuals were sequenced from our samples. One species is represented by 14,841 individuals, and corresponds to a super-abundant species of grass fly (Chloropidae: Oscinellinae: *Olcella* sp.). This species was collected year-round and detected in 11 out of 13 Level III ecoregions across the state; we failed to collect it only in the far northeastern corner of the state, where sampling effort is lowest. We do note that specimens, particularly those from Snapshot Sites, were sorted prior to sequencing to reduce redundancy; this would particularly impact abundance estimates for eusocial insects (honeybees and ants) and collembolans.

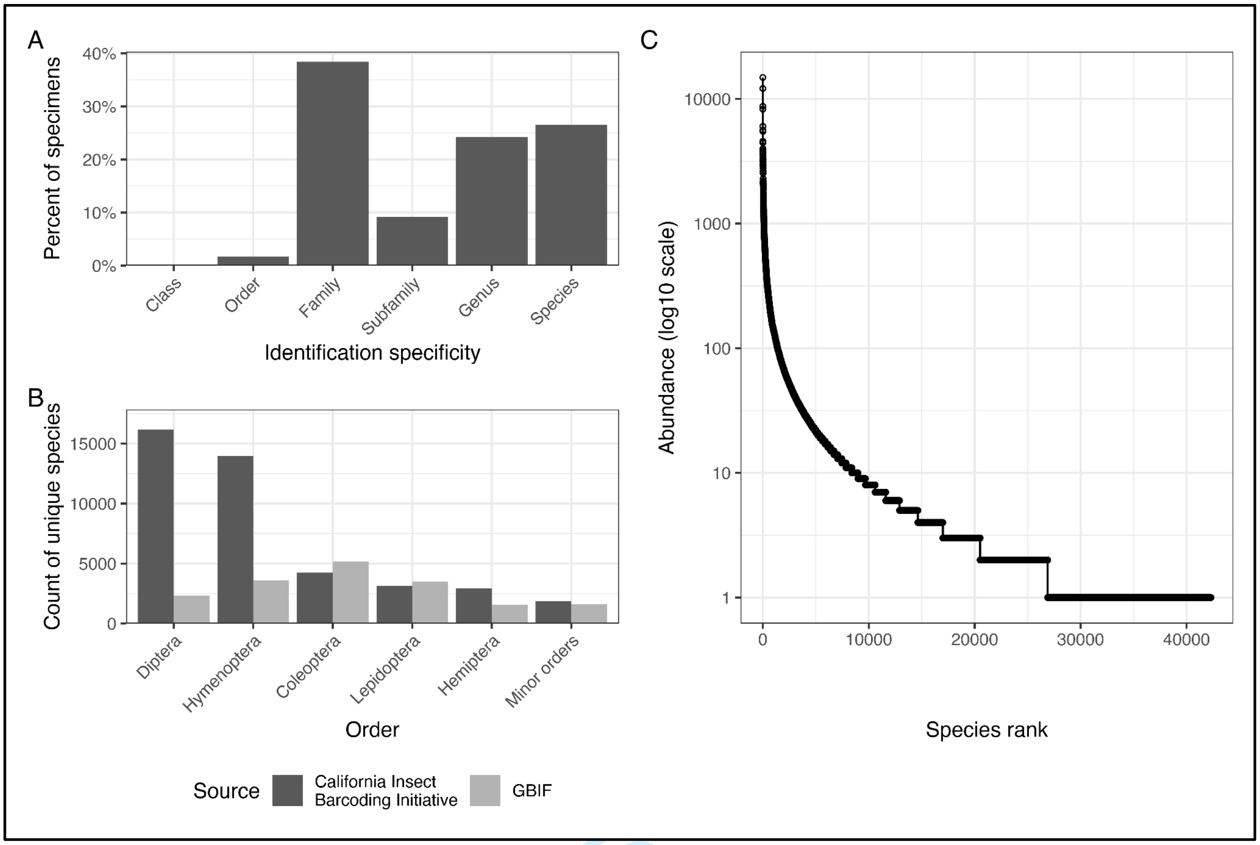

Figure 1: Taxonomic and abundance statistics for California Insect Barcoding Initiative data. Panel A, the taxonomic specificity of identifications assigned to specimens. Panel B, for our dataset (dark gray) and for California insect data derived from GBIF (light grey), the number of unique species for each of the 5 most abundant orders of insects and for the minor orders in aggregate (those with fewer than 2,400 species in our dataset). Panel C, rank-abundance curve for species identified in this project; the y-axis presents the number of individual specimens collected for each species, and the x-axis presents the ordinal rank for each species from most to least abundant.

Impact of collecting method

The composition of sampled specimens is likely shaped by our sampling methods, which were dominated by Malaise traps. For samples sequenced as part of this dataset, our Malaise traps ran for over 170,000 trap hours, and this collecting effort produced 91.6% of all sequenced specimens. All other collecting techniques are responsible for just 8.4% of the total number of specimens (primarily vegetation sweeping, 5.0% of all specimens). However, these collecting techniques captured a disproportionately large fraction of unique species: 13.8% of species were collected only via non-Malaise methods, while 64.4% of species were collected only by Malaise traps, and 20.8% of species were collected using both approaches (Supplementary File 2).

The specimens collected via non-Malaise trap methods display a different taxonomic composition than do specimens collected in Malaise traps. Of the most abundant orders, Hemiptera make up a much higher proportion of non-Malaise trap specimens than Malaise trap specimens (19.9% vs. 7.1%), as do Coleoptera (10.2% vs. 4.0%) and Hymenoptera (20.0% vs. 14.9%) (Supplementary File 2). Nearly all minor orders were more abundant among non-Malaise trap specimens. The non-Malaise methods also yielded specimens for two flightless groups that were never collected in Malaise traps, Acerentomata and Siphonaptera.

Seasonal patterns in collecting

The seasonal distribution of collecting events was fairly even across spring (March–May: 29.9%), summer (June–August: 37.3%), and fall (September–November: 27.9%), but lower in winter (December–February: 4.8%). In contrast, the seasonal distribution of specimens sequenced is biased towards the fall (63.2%) relative to spring (17.1%), summer (19.1%), and winter (0.5%), with a larger proportion of sequenced specimens being caught during fall in the Observatory Site Malaise traps.

Collecting biases across ecoregions

We sampled widely across California, sampling all Level III regions and 80% of Level IV regions by the end of 2024 (Fig. 2A). Ecoregions varied in the number of species collected and sequenced, from 10,585 in the Southern California Mountains surrounding Los Angeles and San Diego, to 625 in the Northern Basin and Range (Fig. 2B). While this variation could reflect true differences in species richness, it may also reflect biases in collecting effort. To assess these potential biases, we developed a metric that measures collection effort relative to the area of each ecoregion (Fig. 2C). In this way we show that the Southern California/Northern Baja Coast is relatively over-collected, likely because three major institutional participants (NHMLAC, UCR, and SDNHM) are present in this region. In contrast, the Central California Valley, for example, is relatively under-collected. This bias likely results from intense agricultural development in the Central Valley. As a result, the majority of the land tenure is private, making the Central Valley the most difficult ecoregion in which to find natural areas to sample, despite its large size.

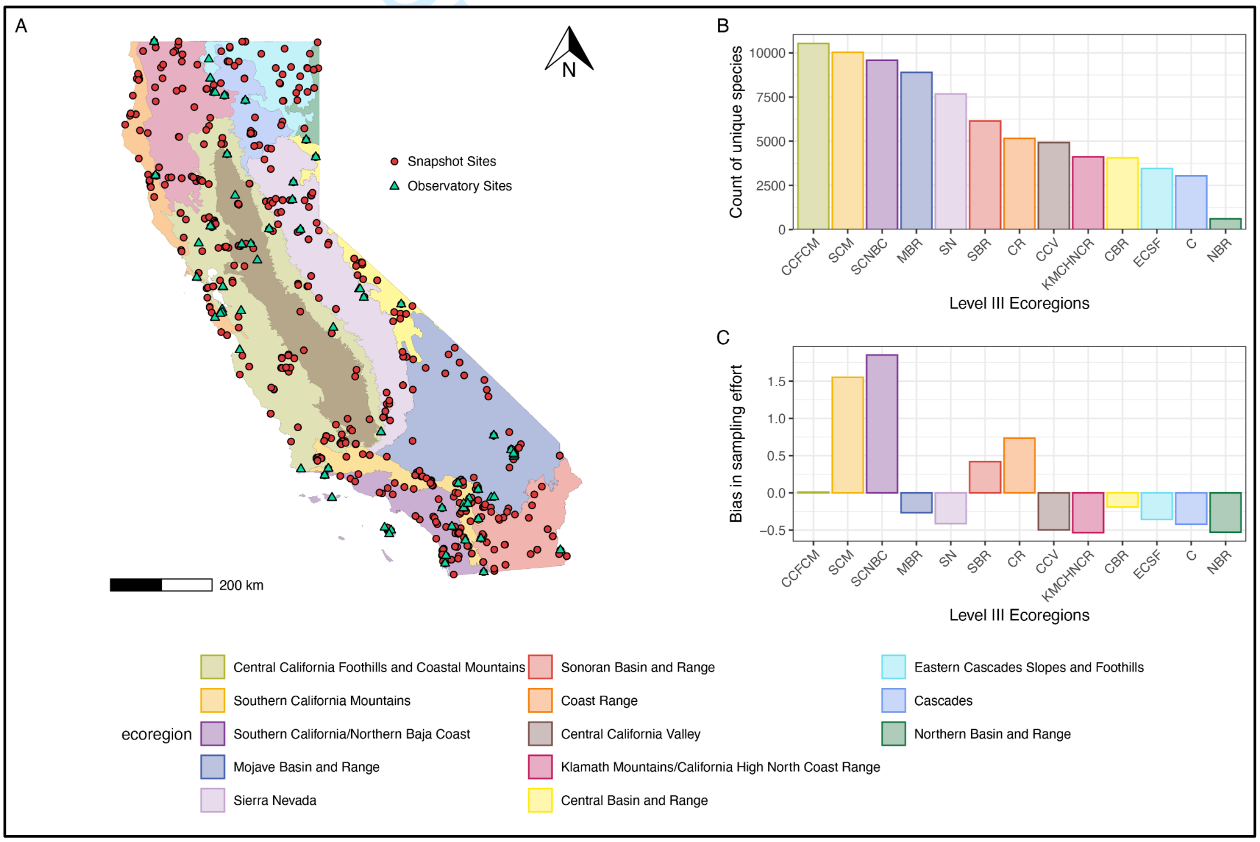

Figure 2: Collecting effort across California. Panel A, distribution of sampling events across California's EPA Level III ecoregions, with blue triangles indicating Observatory Sites and red circles indicating Snapshot Sites. Panel B, the total number of unique species collected per ecoregion. Panel C, the relative collection effort per ecoregion corrected by the relative area of each ecoregion, where values greater than zero represent areas that are over-collected, and values below zero represent areas that are under-collected; values = (EC/TC)/(EA/TA); EC = ecoregion collecting events; TC = total collecting events; EA = ecoregion area; TA = total area.

Predicted biodiversity and per-site sampling completeness

Based on the asymptote obtained using rarefaction/extrapolation with the Chao1 estimator, the estimated lower bound for the number of species present in California is approximately 60,945 (CI: 60,339 - 61,551). Using the iChao1 estimator for all sequenced specimens, the estimated lower bound given our current sampling is approximately 64,915 species. In contrast, using the iChao estimator with only Observatory Site specimens produces an estimated lower bound of approximately 49,535 Malaise-trappable species in California, given our current sampling. Neither the species accumulation curves nor the rarefaction curves yet reach asymptotes (Fig. 3A).

We also inferred per-site sampling completeness for the 17 best-sampled Observatory Sites (Fig. 3B and C). Estimated completeness ranged from 40.7% to 64.6%, with a mean of 53.4%. Because our samples were largely collected via Malaise traps, which undersample non-flying insects, we note that these diversity metrics are underestimates of total insect species richness at these sites.

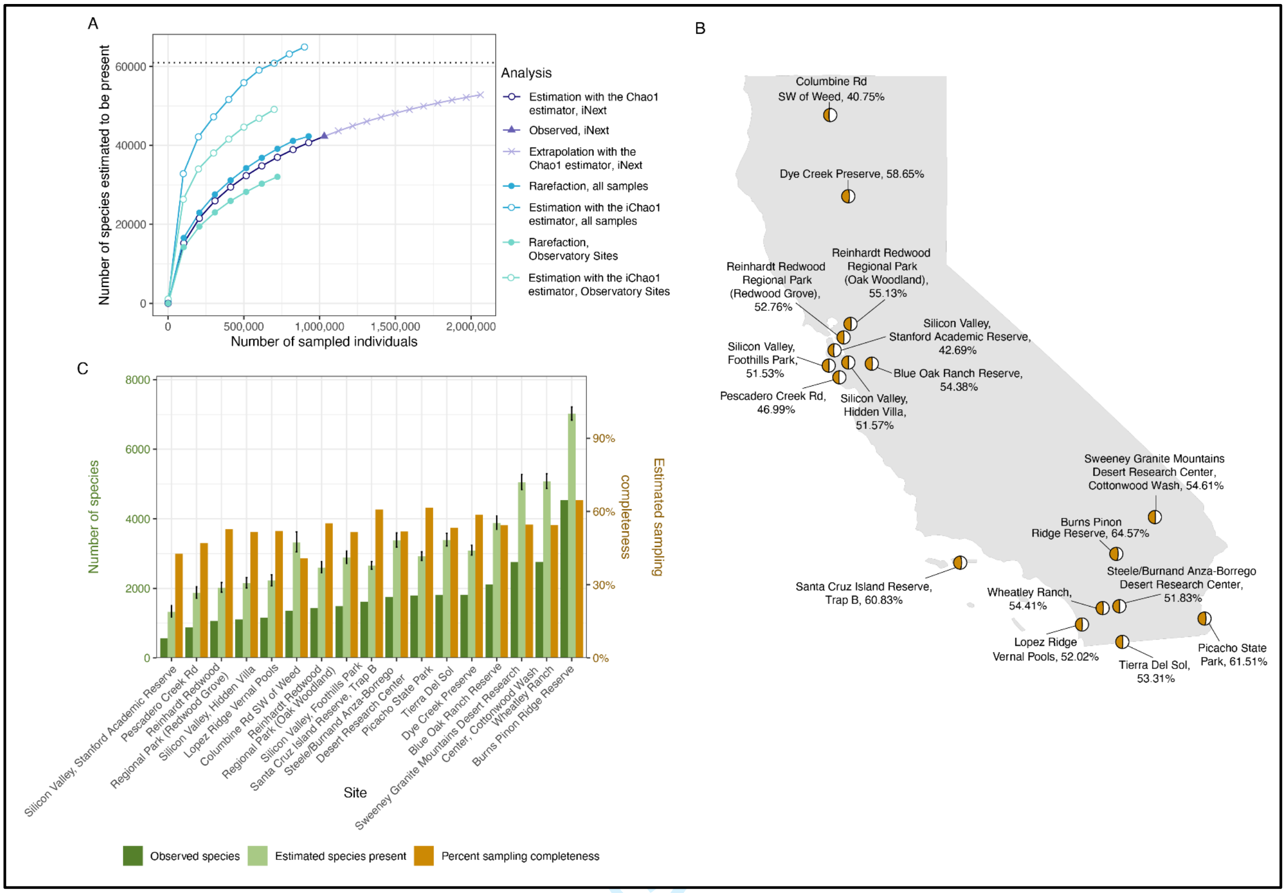

Figure 3: Species richness and sampling completeness estimates. Panel A, Rarefaction and species estimate curves for three methods used to estimate statewide sampling completeness and a lower bound for the number of species present in the state of California. Analysis of all samples using the Chao1 estimator implemented in iNext are presented in purple; analysis of all samples using rarefaction or estimation with the iChao1 estimator presented in blue; and analysis of Observatory Site samples using rarefaction or estimation with the iChao1 estimator presented in teal. The dashed line indicates the asymptote value inferred using the Chao1 estimator rarefaction/extrapolation analysis. Panel B, approximate locations of the 17 best-sampled Observatory Sites, with estimated percent sampling completeness presented as moon plots. Panel C, counts for the actual number of unique observed species (dark green bars), estimated number of species present (light green bars), and percent sampling completeness (orange bars) for each of the 17 sites.

Site and ecoregion dissimilarity

Our data indicate that California's EPA Level III ecoregions host highly dissimilar insect communities (Fig. 4A), with a mean pairwise Bray-Curtis dissimilarity index of 0.94 (range: 0.92-0.99). In the context of this high between-ecoregion dissimilarity, the Klamath Mountains/High North Coast Range and the Cascades are more similar to one another, suggesting a somewhat shared insect fauna of the mountains of Northern California. The Sonoran Basin and Range and the Mojave Basin and Range are also more similar, indicating that these desert ecoregions share somewhat similar insect communities. Nevertheless, we emphasize that all ecoregions host highly dissimilar insect communities.

In general, Observatory Sites within the same ecoregion are more similar to one another than are sites in adjacent ecoregions, and sites in adjacent ecoregions are in turn more similar than sites in nonadjacent ecoregions (Fig. 4B). Observatory Sites within the same ecoregion have a mean estimated dissimilarity of 0.96 (standard error = 0.13; 0.95, 0.97 credible interval). Observatory Sites within adjacent ecoregions have a mean estimated dissimilarity of 0.97 (standard error = 0.01; 0.96, 0.98 credible interval). Observatory Sites within nonadjacent ecoregions have a mean estimated dissimilarity of 0.98 (standard error = 0.01; 0.97, 0.98 credible interval). The standard deviation captured by the combination of month and site random effects is 0.82 (0.71, 0.96 credible interval), indicating that a meaningful amount of variation in dissimilarity is explained by site identity and month of sampling. All R-hat values are equal to 1.

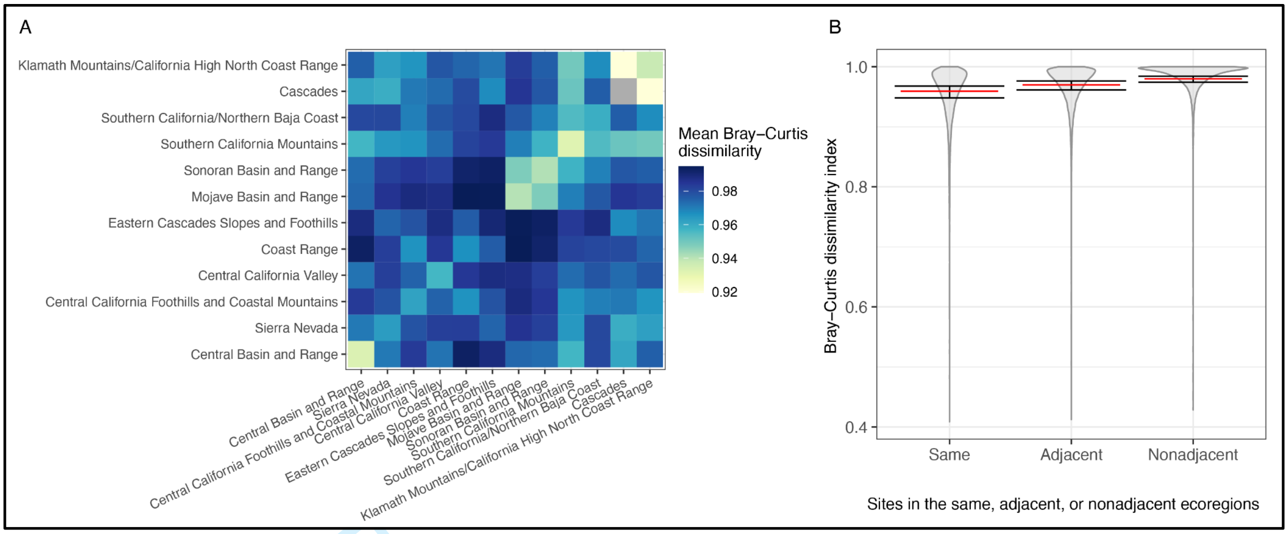

Figure 4: Insect community dissimilarity across ecoregions and Observatory Sites. Panel A, heatmap of average site pairwise Bray-Curtis dissimilarity index for each pair of California EPA Level III ecoregions (excluding the Northern Basin and Range, where we do not have any Observatory Site locations). Darker blue colors, corresponding to higher values of the Bray-Curtis dissimilarity index, indicate less similar insect communities, while lighter green and yellow colors indicate more similar insect communities. Panel B, pairwise Bray-Curtis dissimilarity index for sites within the same ecoregion, within adjacent ecoregions, and within nonadjacent ecoregions; red bars indicate the estimated mean value from a Bayesian zero/one inflated regression for each site pair category, and black crossbars indicate the corresponding credible interval.

The spatial distribution of insect biodiversity in California

Our potential biodiversity heatmap (Fig. 5) identifies areas of particularly high insect species richness in several Level IV ecoregions within the Southern California Mountains, including the Arid Montane Footslopes, Southern California Montane Conifer Forest, and Southern California Lower Montane Shrub and Woodland. Each of these regions contains one-square-kilometer areas that are predicted to contain over 8,000 insect species. Additionally, the Mojave Basin and Range contains Level IV ecoregions with high predicted diversity, including the Eastern Mojave Low Ranges and Arid Footslopes and the Eastern Mojave Basins (also predicted to have over 8,000 insect species).

Our diversity predictions only take into account sampled species, and are therefore influenced by sampling effort. Thus, while we infer high species richness in the Southern California Mountains, this ecoregion is also highly over-collected in our dataset (Fig. 2C). However, the Mojave Basin and Range, another area of high inferred insect diversity, is actually undercollected, suggesting that this region contains true peaks in diversity.

The effect of sampling effort can also be seen at smaller spatial scales. For example, the Burns Piñon Ridge Reserve, which includes one of the most heavily sampled Observatory Sites, has a predicted richness value of 7,092 species based on species collected both at that site and other sites with the same vegetation types and in the same ecoregion. Given that our predicted sampling completeness for the Observatory Site Malaise trap at that location is 64.57%, this result suggests that there may be as many as 10,983 insect species at that location.

Our interpolated diversity maps also highlight remaining gaps in sampling, corresponding to vegetation type-ecoregion combinations from which we do not yet have sequenced specimens. Some of these areas, indicated in gray, have been sampled but specimens have not yet been sequenced and therefore cannot be included in the present analysis. Many sampling gaps fall in highly agriculturized Central Valley ecoregions, where collecting is challenging.

The patterns that we observe for all insects are largely mirrored at the Order level, particularly for groups with sufficient sampling and species diversity to yield informative richness indices (i.e. Coleoptera, Diptera, Hemiptera, Hymenoptera, Lepidoptera, Neuroptera, Orthoptera, and Thysanoptera) (Supplementary File 1).

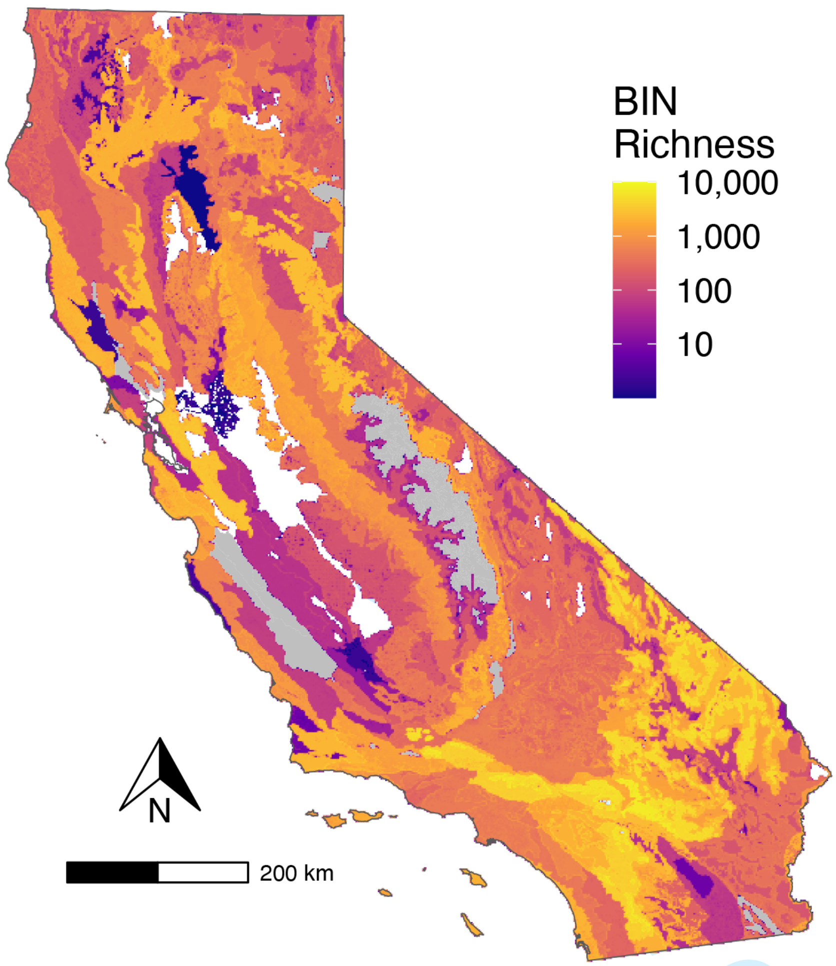

Figure 5: Geographic distribution of potential richness estimate for insects across California. Estimates are based on rule-based spatial range interpolation constrained by ecoregion and vegetation type for every species in the dataset. The number of species predicted in each square kilometer is represented by the color plotted on the map, with lighter colors indicating higher species richness (color scale log-transformed), gray indicating EPA Level IV ecoregions that have been sampled but not yet sequenced, and white indicating ecoregions and EVT cover type combinations for which we lack insect samples.

Discussion

The California Insect Barcoding Initiative represents one of the largest efforts ever made to document regional insect biodiversity using DNA barcoding, with 1,030,728 hexapod specimen records to date. We maximized the biodiversity collected by sampling in every EPA Level III ecoregion across California. This effort resulted in 42,305 unique species collected and sequenced, as delimited by DNA barcode-based molecular operational taxonomic units. Using these data, we infer statewide patterns of insect biodiversity and demonstrate that California's ecoregions harbor strongly dissimilar insect communities, with overall diversity peaking in the Southern California Mountains ecoregion and the Mojave Basin and Range ecoregion. The barcode sequences generated by this project also represent a major contribution to the development of a reference library that can be used for future rapid and affordable species identifications.

Collecting methodology strongly impacts the results of insect surveys, as different methods preferentially target different taxa (Anderson *et al*., 2024; Souto-Vilarós *et al*., 2025). Like other large-scale insect surveys (Hebert *et al*., 2016; Karlsson *et al*., 2020a, 2020b), the California Insect Barcoding Initiative heavily relied on Malaise trapping. While Malaise traps are easily deployed and allow consistent passive collection, they do not effectively sample groups that do not fly frequently or at all (e.g., many beetles, springtails, aquatic insects, scale insects) or insects that are able to see and avoid traps (e.g., dragonflies, butterflies, etc.) (Devigne & De Biseau, 2014; Lamarre *et al*., 2012; Montgomery *et al*., 2021; Souto-Vilarós *et al*., 2025; Srivathsan *et al.*, 2023). Our extensive use of Malaise traps did significantly influence the composition of collected specimens. However, to mitigate these sampling biases, we employed a variety of additional collecting methods which increased the diversity of insects collected. This approach has generated a highly biodiverse dataset, but one that is also more heterogeneous compared to those that target a single community or use a single collecting method (e.g., Caterino & Recuero, 2023).

Our study also demonstrates the utility of using DNA barcodes to understand insect biodiversity at large scales. Compared to projects that rely on morphological identification, such as the Swedish Malaise Trap Project (Karlsson *et al*., 2020a, 2020b), the California Insect Barcoding Initiative was able to rapidly sequence, database, and delimit over 890,000 specimens in three years. Barcoding is also advantageous in that it permits the identification of cryptic species. Since cryptic species appear to comprise a substantial fraction of total insect biodiversity (e.g., Li and Wiens, 2023; Basset *et al*., 2025), it is critical that surveys of insect fauna utilize species identification approaches that are capable of detecting these species. The main disadvantage of barcoding is that specimens are delimited to MOTUs, rather than named species. However, ongoing efforts to generate barcodes from authoritatively identified specimens that are vouchered in legacy collections will facilitate more precise, traditional species identifications, as will additional rigorous taxonomic work.

California is tremendously biodiverse (Calsbeek *et al*., 2003; Myers *et al*., 2000), and despite decades of effort we are just now beginning to quantify the state's insect diversity. The California Insect Barcoding Initiative builds upon previous statewide surveys led by the University of California in the 1940s–1980s and between 2015–2020. These efforts collectively described approximately 12% of California's insect diversity, resulting in estimates that there are 26,000–100,000 insect species present in the state, of which 6% were considered new to science (University of California, Davis, 2022).

Estimates of the number of insect species present in the state can also be produced by leveraging digitally-available legacy specimens. For instance, the GBIF database contains 17,584 insect species represented by specimens collected in California, less than half the number of species collected and sequenced by the California Insect Barcoding Initiative. Discrepancies are particularly pronounced in some groups: for instance, our dataset contains 688% more Diptera species and 387% more Hymenoptera species for California than does GBIF. These differences likely reflect both our use of DNA barcoding for species delimitation, allowing us to more easily make species identifications, detect cryptic lineages, and detect and distinguish previously undescribed species (Meier *et al*., 2025) and the limited proportion of North American collections holdings that are currently digitized (ca. 5% of pinned insect specimens) and thus discoverable via GBIF (Cobb *et al*., 2019).

Despite several years of significant sampling effort statewide, rarefaction curves demonstrate that we have not yet sampled every insect species found in California. We therefore estimate the likely number of insect species present in the state, inferring a conservative lower bound of approximately 61,000 species in California. Our analyses further indicate that the project's sampling completeness is 65.2–69.4%. While 61,000 species is a large number, we expect this is an underestimate since it is heavily based on sampling done with Malaise traps, which undersample many parts of the insect fauna. Furthermore, our extrapolation methods are best suited to collecting methods that are not biased towards rare or abundant taxa (Chao *et al*. 2009). However, to present a more comprehensive view of the potential insect biodiversity of California, we decided to include specimens collected via all sampling methods, including those which preferentially targeted uncommon species.

Given these caveats, the true number of species present in California is likely to be well above the lower bound of 61,000, although the degree to which our current prediction is an underestimate is difficult to ascertain. Nevertheless, it is clear that our sampling has not yet fully captured the total number of insects present in the state. This pattern is not atypical for insect biodiversity studies. For example, in an ongoing long-term insect survey of Costa Rica's Area de Conservación Guanacaste (Quicke *et al*., 2024), species accumulation curves have not yet reached an asymptote. This begs the question: if more than 40 years of continuous sampling have been insufficient to collect every insect species in a 147,000 hectare area of Costa Rica, how much sampling will be needed for the 42 million hectare state of California?

The California Insect Barcoding Initiative also sheds light on the potential spatial distribution of insect biodiversity. Across the state, we find that insect species richness appears to peak in the Southern California Mountains and in the Mojave Basin and Range ecoregions. Biases in collecting effort may have inflated biodiversity estimates in the Southern California Mountains. However, the Mojave Basin and Range is actually under-sampled relative to the area of the ecoregion, and thus likely represents a true peak in insect biodiversity. This prediction for all insects matches the pattern observed for bee species richness, which peaks in xeric-temperate areas around the world (Orr *et al.*, 2021).

On the whole, we find that sites across the state host strongly dissimilar insect communities. Community dissimilarity is high both within and between ecoregions, although ecoregions in closer spatial proximity are slightly more similar. Interestingly, two ecoregions which encompass Northern California mountain ranges, the Cascades ecoregion and the Klamath Mountains/California High North Coast Range ecoregion, are the two most similar ecoregions in terms of their insect fauna. These two ecoregions share many similarities in tree diversity, elevation, and climate (Griffith *et al.*, 2016), in addition to being adjacent to one another. Likewise, the state's two desert ecoregions, the Mojave Basin and Range and the Sonoran Basin and Range, are more similar to one another than to other ecoregions, and are highly dissimilar to all other ecoregions in the state. This pattern suggests that the warm, arid, and strongly seasonal abiotic conditions in these deserts drive common patterns of insect community composition.

To date, our understanding of the spatial distribution of biodiversity across California has largely been based on five groups of organisms: birds, mammals, amphibians, reptiles, and plants (California Department of Fish and Wildlife, 2018). Data for these groups are compiled in the California Department of Fish and Wildlife's Areas of Conservation Emphasis Terrestrial Native Species Richness maps, which are used to identify biodiversity hotspots and areas for future conservation under the state's 30x30 initiative. However, none of these taxa, individually or in aggregate, exhibit the same pattern of diversity as the insects. For instance, in some areas of the Mojave Basin and Range where we find high values of insect diversity, the ACE

Terrestrial Native Species Richness Summary reports diversity values that fall in the lowest quintile statewide. Thus, because vertebrates and plants appear to be poor proxies for insects (and likely other invertebrates), a failure to explicitly consider insects in spatial conservation planning will result in a failure to protect this important component of terrestrial biodiversity.

The data generated via the California Insect Barcoding Initiative have substantial applications beyond this study. As discussed above, they have important implications for state and regional conservation planning. Moreover, our efforts are contributing to the generation of a robust DNA barcode reference library for insects in the state of California. This reference library will facilitate future investigation of insect diversity in the region; in particular, it has the potential to greatly improve the efficacy of environmental DNA (eDNA) survey efforts, wherein DNA present in environmental samples is collected and sequenced in lieu of directly sampling target organisms. Successful taxonomic identifications in eDNA metabarcoding analyses depend on the availability of reference sequences, which currently have poor taxonomic resolution for most insect groups (Leandro *et al*., 2024). Thus, by greatly increasing the number of reference sequences available for California insects, the California Insect Barcoding Initiative will improve the accuracy of future eDNA surveys in the region.

Moreover, the abundance and diversity data collected by the Observatory Sites permit phenological inferences and can be used as a starting point for long term population or community monitoring. The exact location and time at which each standardized Malaise trap was deployed is recorded, and many Observatory Sites are paired with weather stations from which abiotic data can be derived. Moreover, the wet-weight biomass and taxonomic composition of the Observatory Site samples can be directly compared to future replicate samples to understand changes in insect communities over time. In the face of current insect declines and the increased anthropogenic movement of species, establishing insect diversity, abundance, and distribution baselines now is more important than ever.

In addition to producing a resource to enable future biodiversity studies, this dataset has great utility for detecting both native species and species not previously known from California and can be used as a starting point for tracking the distributions and movement of invasive species. For example, we detected an Asian aphid species, *Shinjia orientalis* (Mordvilko); an aphid species native to the eastern United States, *Hyperomyzus picridis* (Börner & Blunk); a Floridian fire ant, *Solenopsis abdita* Thompson; and a cricket known from the southeastern United States, *Gryllus rubens* (Scudder). None of these species were previously known to occur in California. For several minor orders, we actually find that the most abundant species is introduced. For example, the European earwig, *Forficula auricularia* L., is by far the most abundant dermapteran; the black webspinner *Oligotoma nigr*a Hagen, which is originally from India, is the most abundant embiopteran; and the cosmopolitan dimorphic springtail, *Entomobrya atrocincta* Schott, is the most abundant entomobryomorphan Collembola.

The California Insect Barcoding Initiative has made significant progress towards creating a comprehensive DNA barcode reference library for the insect fauna of California, but we have a long way to go before this goal is fully met. For instance, the current dataset contains sampling gaps, both taxonomic and geographic. These gaps continue to be filled by additional targeted collecting throughout 2025. However, as sampling progresses, an increasingly large fraction of the remaining unsampled species will be rare taxa. These species will likely require large amounts of collecting and/or the use more labor-intensive collecting methods to sample.

In addition, fully identifying sequenced specimens to named species, rather than to BINs, will be a substantial and ongoing challenge. Barcode sequencing of expertly-identified specimens from legacy collections will help ameliorate this challenge; however, the large number of undescribed and cryptic species that have been discovered as part of this survey cannot be identified in this manner. If those specimens are to ever have species identifications, then we must prioritize investing resources into the advancement of basic taxonomic revisionary science. These next steps will be difficult but will greatly improve our understanding of California's diverse insect fauna.

Conclusion

The California Insect Barcoding Initiative represents one of the most extensive DNA barcoding efforts yet undertaken for regional insect biodiversity, and demonstrates that high-throughput sequencing can transform our understanding of macroecological patterns. By generating more than one million specimen records, we show that insect richness in California is far greater than previously documented and that spatial patterns of diversity emerge which have not been captured in broad-scale biodiversity assessments of vertebrate and plant taxa.

Beyond establishing a foundation for a scalable reference library, our results underscore the power of DNA barcoding to detect cryptic diversity, reveal novel biogeographic patterns, and provide baselines for monitoring community change in the face of global insect declines. While taxonomic resolution and sampling completeness remain challenges, these limitations highlight the urgent need to integrate molecular inventories with revisionary taxonomy and targeted sampling. Continued development of this and similar resources will not only refine estimates of species richness and distributions, but also contribute to understanding the drivers of insect diversity and to strengthening the role of invertebrates in conservation planning.

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Supplementary Material

Supplementary Table 1: CIBI Collecting permits and authorizations: List of permits and authorizations used by institutions and partners for collecting insects.

Supplementary Table 2: Observatory sites: Tab 1: List of Observatory Site locations with the number of samples collected and the number sequenced. Tab 2: Wet weight biomass measurements for individual Observatory Site samples.

Supplementary Table 3: CIBI insect data from BOLD: Specimen records with taxonomic and collection information downloaded from boldsystems.org on June 25, 2025.

Supplementary Table 4: GBIF data: Compiled GBIF datasheets for insects in California downloaded on August 21, 2025.

Supplementary File 1: Species richness Order maps: Species richness maps generated for each individual hexapod order.

Supplementary File 2: Collecting methods: Comparison of the number of BINs collected with Malaise traps and non-Malaise methods.

Supplementary File 3: BIN counts: Count of specimens sequenced for each BIN.

Supplementary File 4: Non-log transformed species richness map: Species richness map showing non-log transformed data.